

实验4: CacheLab

1.实验准备

```
racalia202212063020@THESEI × + v
racalia202212063020@THESELECTED3:~/cachelab-handout$ ls
Makefile  cachelab.c  csim-ref  driver.py  test-trans.c  traces
README    cachelab.h  csim.c    test-csim  tracegen.c    trans.c
racalia202212063020@THESELECTED3:~/cachelab-handout$ |
```

2.根据PPT写出实验文件

```
#include<fcntl.h>
#include<getopt.h>
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
#include"cachelab.h"

void usage(void){exit(1);}
int verbose = 0,s,E,b,S,B;

int T=0;
typedef __uint64_t uint64_t;

typedef struct lineNode{
    int t;
    uint64_t tag;
} * groupNode;

enum Category{ HIT,MISS,EVICTION};
unsigned int result[3];
const char *categoryString[3]={"hit","miss","eviction"};
groupNode *cache;
void init();
void destory();
FILE *opt(int argc,char **argv);
void findCache(uint64_t tag,int group_pos,char *result);

int main(int argc,char **argv)
{
```

```

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void findCache(uint64_t tag,int group_pos,char *result);

int main(int argc,char **argv)
{

```

```

        ++result[category];
        group[pos].tag=tag;
        group[pos].t=T;
        if(verbose) strcat(resultV,categoryString[category]);
    }
}

void findCache(uint64_t tag,int group_pos,char *resultV)
{
    groupNode group =cache[group_pos];
    int min_t_pos=0,empty_line=-1;
    for(int i=0;i<E;++i){
        struct lineNode line=group[i];
        if(!line.t)
            empty_line=i;
        else{
            if(line.tag==tag){
                setResult(group,HIT,tag,i,resultV);
                return;
            }
            if(group[min_t_pos].t>line.t)
                min_t_pos=i;
        }
    }
    setResult(group,MISS,tag,empty_line,resultV);
    if(empty_line==-1)
        setResult(group,EVICTION,tag,min_t_pos,resultV);
}

void destory()
{
    for(int i=0;i<S;++i){

```

```

        free(cache[i]);
    }
    free(cache);
}

FILE *opt(int argc,char **argv)

FILE *tracefile;
for(int c;(c=getopt(argc,argv,"hvsEbt"))!=EOF;){
    switch(c){
        case 'h':
            usage();
            break;
        case 'v':
            verbose=1;
            break;
        case 't':
            tracefile=fopen(argv[optind],"r");
            if(tracefile==NULL) usage();
            break;
        case 's':
            s=atoi(argv[optind]);
            if(s<=0) usage();
            S=1<<s;
            break;
        case 'E':
            E=atoi(argv[optind]);
            if(E<=0) usage();
            break;
        case 'b':

```

3.测试击中数量

```
L 10,1 miss
M 20,1 miss hit
L 22,1 hit
S 18,1 hit
L 110,1 miss eviction
L 210,1 miss eviction
M 12,1 miss eviction hit
hits:4 misses:5 evictions:3
```

4.观察得分

Points (s,E,b)	Hits	Your simulator		Reference simulator			
		Misses	Evicts	Hits	Misses	Evicts	
3 (1,1,1)	9	8	6	9	8	6	traces/yi2
3 (4,2,4)	4	5	2	4	5	2	traces/yi.
3 (2,1,4)	2	3	1	2	3	1	traces/dav
3 (2,1,3)	167	71	67	167	71	67	traces/tra
3 (2,2,3)	201	37	29	201	37	29	traces/tra
3 (2,4,3)	212	26	10	212	26	10	traces/tra
3 (5,1,5)	231	7	0	231	7	0	traces/tra
6 (5,1,5)	265189	21775	21743	265189	21775	21743	traces/lon